Learned Target Assignment for Synthetic Tabular Data

Relabeling synthetic data to outperform original training data.

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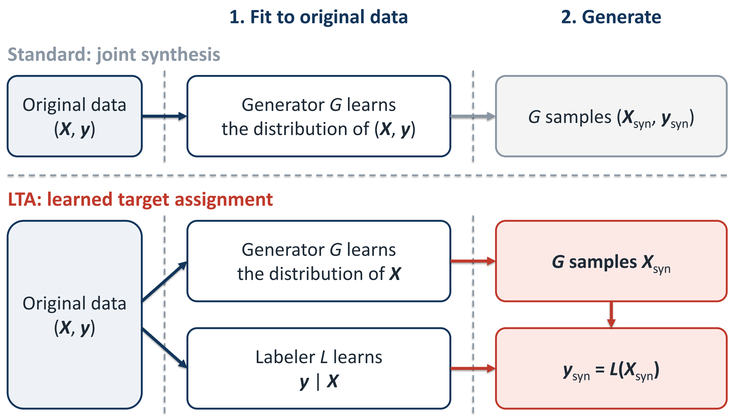
Why generate *X* and *y* separately?

A supervised ML problem has input features ***X*** and a target ***y***. Synthetic data helps a downstream model when it preserves the dependency of ***y*** on ***X***.

CTGAN, **TVAE** and **TabDDPM** model (***X***, ***y***) jointly, unaware of the target: ***y*** is one more feature column, and nothing ties a generated ***y*** to its ***X***.

LTA models this dependency directly: a generator *G* samples the features, and a labeler *L* fitted to original (***X***, ***y***) assigns their targets.

Method



- *G* can be any tabular generator and *L* any supervised model.
- Both are fitted to original training rows only.

Evaluation

- **Data:** Adult, Census KDD, Covertype, MNIST12, MNIST28 (macro F1); California Housing, News (R^2).
- **Models:** *G* = CTGAN, TVAE or TabDDPM (100,000 rows); *L* = RF, XGB or DNN.
- **Scoring:** each training set trains a fresh neural network, which is tested on held-out original rows.
- **Training sets:** original data only, or synthetic data with generated ***y*** (standard) or $\boldsymbol{y}_{syn} = L(\boldsymbol{X}_{syn})$ (LTA), alone or added to the original data.
- **Ablation:** *L* relabels the $\boldsymbol{X}_{syn}$ of a joint *G* (right). 2 seeds; 588 runs.

Takeaways

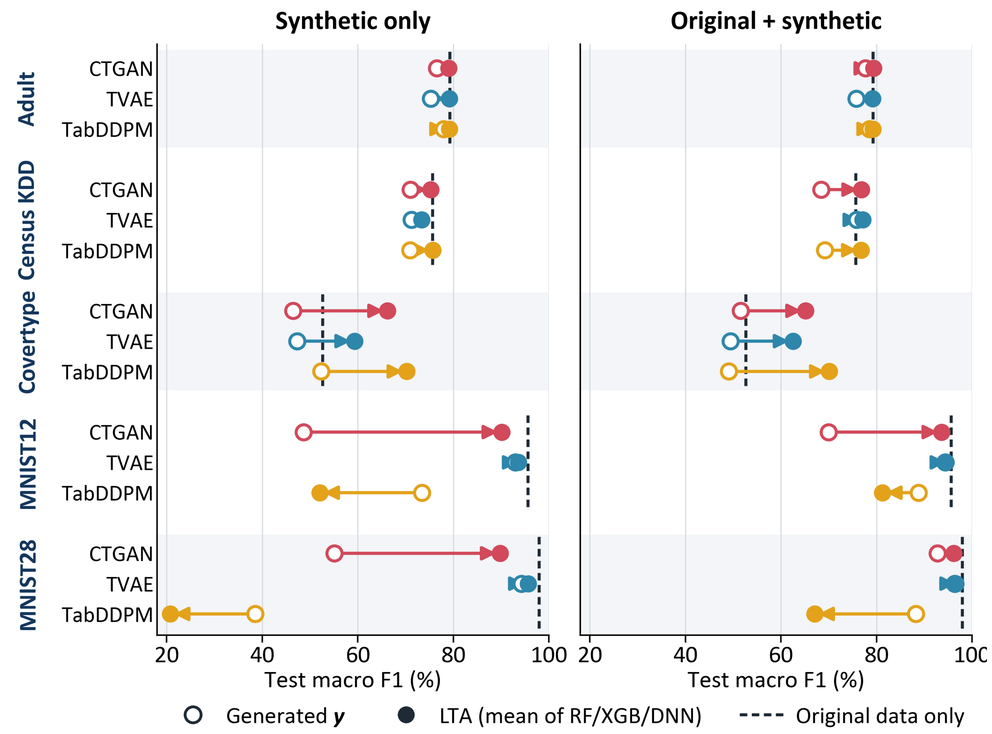
- Labeling synthetic features with a model trained on original data is a simple add-on to any generator.
- Exception: TabDDPM on MNIST. Its digits drift from the original ones, so a labeler trained on original data mislabels them.

References

1. Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. Modeling tabular data using conditional GAN. NeurIPS 2019. (CTGAN, TVAE)
2. Kotelnikov, A., Baranchuk, D., Rubachev, I., and Babenko, A. TabDDPM: Modelling tabular data with diffusion models. ICML 2023.
3. Breiman, L. Random forests. Machine Learning 45, 2001.
4. Chen, T. and Guestrin, C. XGBoost: A scalable tree boosting system. KDD 2016.

Result: synthetic data that rivals original data

Generator	Classification: macro F1 (%)				Housing: R^2			
	Synthetic only		Original + synthetic		Synthetic only		Original + synthetic	
	Generated ***y***	LTA	Generated ***y***	LTA	Generated ***y***	LTA	Generated ***y***	LTA
CTGAN	59.5	**80.1**	72.1	**82.2**	0.63	**0.80**	0.64	**0.80**
TVAE	76.2	**80.2**	78.4	**82.0**	0.71	**0.78**	0.70	**0.78**
TabDDPM	62.6	**59.6**	74.8	**74.9**	0.79	**0.80**	0.79	**0.80**
Original data only	80.2				0.51			



Regression (same layout; dashed line = original data only)

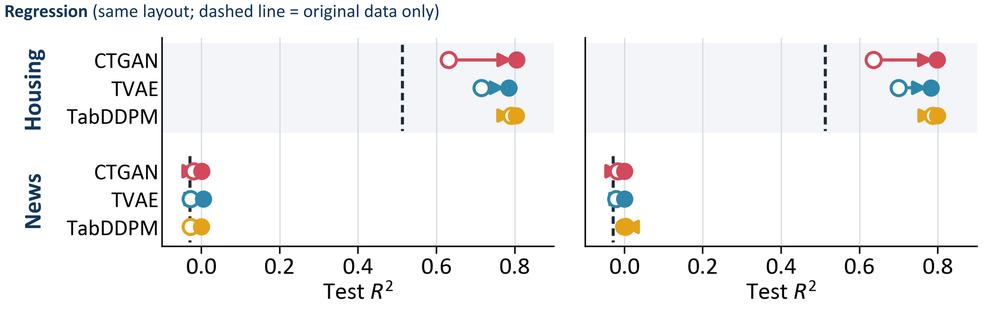
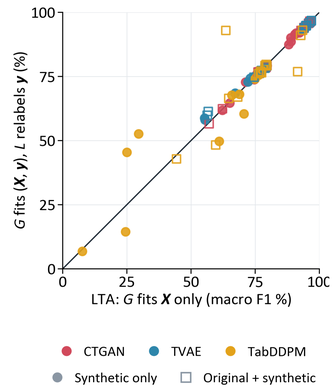


Table and figure: averaged over 2 seeds (LTA also over RF, XGB and DNN); classification cells also average over the 5 tasks. Open circle: the generator's own targets. Filled circle: LTA. Generated targets fall short of original data on 5 of 5 classification tasks for every generator; News has no predictable signal for any method, including original data ($|R^2| \le 0.03$).

Does the generator need *y*?



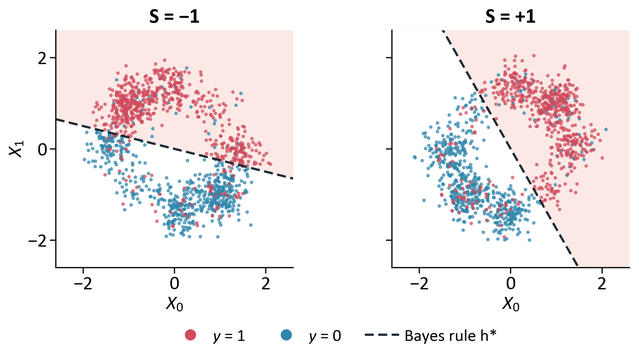
Each point is one task, generator and labeler. On the diagonal, fitting *G* to (***X***, ***y***) instead of ***X*** alone makes no difference.

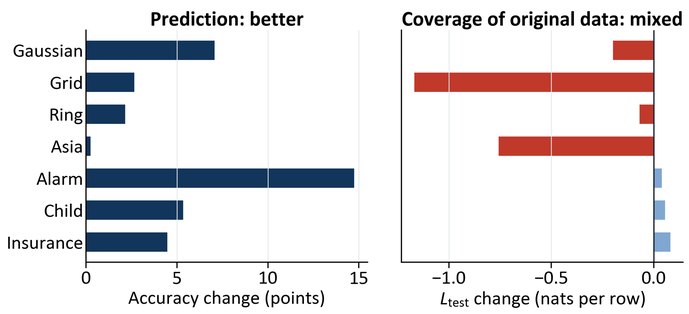
Joint minus LTA: **+0.5 points** (95% CI −3.0 to +3.9; Holm p = 1.00).

Fitting *G* to ***y*** as well adds no detectable benefit.

Simulated data with known truth

Ring: 8 Gaussian clusters in (X_0, X_1). A binary feature *S* sets the slope of the Bayes boundary h^*; 10% of labels are flipped. Six more datasets: Gaussian, Grid and 4 Bayesian networks.





L_{test}: fit a simple density model to the synthetic table, then average its log-density over original test rows. Higher means the synthetic table covers the original data better.

LTA raises accuracy on all 7 datasets (+0.3 to +14.7 points, within 1.2 of original data) but lowers L_{test} on 4: hard labels erase the target's natural randomness.